MAT2040

Tutorial 3

TA Group

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June 12&13, 2024

Quick review

1. Matrix transpose and properties
2. Matrix symmetry
3. Matrix inverse and properties
4. Elementary matrix and its inverse

Question 1

Suppose $\mathbf{a}, \mathbf{b} \in \mathbb{R}^4$, and

$$\mathbf{a}\mathbf{b}^T = \begin{bmatrix} 39 & 99 & 420 & 207 \\ 45 & 33 & 782 & 199 \\ 100 & 80 & 49 & 73 \\ 59 & 18 & 75 & 46 \end{bmatrix},$$

then what is $\mathbf{a}^T\mathbf{b}$?

Solution

$$\mathbf{a}\mathbf{b}^T = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 & a_1 b_4 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 & a_2 b_4 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 & a_3 b_4 \\ a_4 b_1 & a_4 b_2 & a_4 b_3 & a_4 b_4 \end{bmatrix}.$$

$$\mathbf{a}^T\mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4 = 167.$$

Question 2

If $A = A^T$ and $B = B^T$, which of the following matrices are symmetric?

(a) $A^2 - B^2$

(b) $(A + B)(A - B)$

(c) ABA

(d) $ABAB$

Solution

(a)

$$(A^2 - B^2)^T = (A^2)^T - (B^2)^T = (A^T)^2 - (B^T)^2$$

Since $A = A^T$ and $B = B^T$

$$(A^2 - B^2)^T = (A^T)^2 - (B^T)^2 = A^2 - B^2$$

Therefore, the matrix is symmetric.
(b) $((A+B)(A-B))^T = (A-B)^T(A+B)^T = (A^T - B^T)(A^T + B^T) = (A-B)(A+B)$. As $(A+B)(A-B)$ is not always equal to $(A-B)(A+B)$, the matrix is not symmetric. E.g., if $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then $(A+B)(A-B) = \begin{bmatrix} 3 & 11 \\ 5 & 18 \end{bmatrix}$, not symmetric.

Solution

(c)

$$(ABA)^T = A^T B^T A^T = ABA$$

Therefore, the matrix is symmetric.

(d)

$$(ABAB)^T = B^T A^T B^T A^T = BABA$$

As $ABAB \neq BABA$ in general, the matrix is not symmetric. E.g., if $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then $ABAB = \begin{bmatrix} 27 & 27 \\ 54 & 54 \end{bmatrix}$, not symmetric.

Question 3

Find the inverse of A, where

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix},$$

(i) by Gauss elimination;

(ii) by the elementary transformation of block multiplication.

Solution

(i) Using Row transformation:

$$\left[\begin{array}{cc} A & I_4 \end{array}\right] = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & -2 & -1 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & -1 & 0 & 1 & 0 \\ 0 & -2 & -2 & 0 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & -2 & -2 & -1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 2 & 0 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 4 & 1 & -1 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & 0 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 & 0 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & 0 & \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 0 & 1 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix} = \left[\begin{array}{cc} I_4 & A^{-1} \end{array}\right]$$

thus $A^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$.

Solution

(ii) $A = \begin{bmatrix} B & B \\ B & -B \end{bmatrix}$ where $B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$.

$$\begin{bmatrix} I_2 & 0 \\ -I_2 & I_2 \end{bmatrix} \begin{bmatrix} B & B \\ B & -B \end{bmatrix} \begin{bmatrix} I_2 & -I_2 \\ 0 & I_2 \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & -2B \end{bmatrix},$$

thus

$$A = \begin{bmatrix} I_2 & 0 \\ -I_2 & I_2 \end{bmatrix}^{-1} \begin{bmatrix} B & 0 \\ 0 & -2B \end{bmatrix} \begin{bmatrix} I_2 & -I_2 \\ 0 & I_2 \end{bmatrix}^{-1},$$

$$A^{-1} = \begin{bmatrix} I_2 & -I_2 \\ 0 & I_2 \end{bmatrix} \begin{bmatrix} B & 0 \\ 0 & -2B \end{bmatrix}^{-1} \begin{bmatrix} I_2 & 0 \\ -I_2 & I_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}B^{-1} & \frac{1}{2}B^{-1} \\ \frac{1}{2}B^{-1} & -\frac{1}{2}B^{-1} \end{bmatrix}.$$

So we have

$$A^{-1} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}.$$

Question 4

(a) Determine the value(s) of λ for which the matrix $\mathbf{A}$ below is invertible.

$$\begin{bmatrix} 1 & \lambda & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) For those values found in part (a), find the inverse of $\mathbf{A}$.

Solution

(a) The given matrix is reduced as:

$$\begin{bmatrix} 1 & \lambda & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to -R_1 + R_2} \begin{bmatrix} 1 & \lambda & 0 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to -R_3 + R_2} \begin{bmatrix} 1 & \lambda & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A is invertible if and only if $1 - \lambda \neq 0$, i.e., $\lambda \neq 1$

(b) Performing Gauss-Jordan elimination, we have:

$$\left[\mathbf{A} \middle| \mathbf{I}\right] = \left[\begin{array}{ccc|ccc} 1 & \lambda & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to -R_1 + R_2} \left[\begin{array}{ccc|ccc} 1 & \lambda & 0 & 1 & 0 & 0 \\ 0 & 1-\lambda & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_3 + R_2} \left[\begin{array}{ccc|ccc} 1 & \lambda & 0 & 1 & 0 & 0 \\ 0 & 1-\lambda & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{1-\lambda} R_2} \left[\begin{array}{ccc|ccc} 1 & \lambda & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{-1}{1-\lambda} & \frac{1}{1-\lambda} & \frac{-1}{1-\lambda} \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

Solution

$$\xrightarrow{R_1 \to -\lambda R_2 + R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1-\lambda} & \frac{-\lambda}{1-\lambda} & \frac{\lambda}{1-\lambda} \\ 0 & 1 & 0 & \frac{-1}{1-\lambda} & \frac{1}{1-\lambda} & \frac{-1}{1-\lambda} \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

So we have $\mathbf{A}^{-1} = \begin{bmatrix} \frac{1}{1-\lambda} & \frac{-\lambda}{1-\lambda} & \frac{\lambda}{1-\lambda} \\ \frac{-1}{1-\lambda} & \frac{1}{1-\lambda} & \frac{-1}{1-\lambda} \\ 0 & 0 & 1 \end{bmatrix}$, $(\lambda \neq 1)$.